2.67 2D Matrices: Factorisation of Determinants

Worked Solutions

1 a)

$$\begin{aligned}|\mathbf{A}| &= \begin{vmatrix} 4 & 40 \\ 12 & 600 \end{vmatrix} \\ &= 4 \times 600 - 40 \times 12 \\ &= 2400 - 480 \\ &= 1920\end{aligned}$$

1 b)

$$\begin{aligned}&10 \begin{vmatrix} 4 & 4 \\ 12 & 60 \end{vmatrix} \\ &= 10(4 \times 60 - 4 \times 12) \\ &= 10(240 - 48) \\ &= 10 \times 192 \\ &= 1920\end{aligned}$$

1 c)

$$\begin{aligned}&40 \begin{vmatrix} 1 & 1 \\ 12 & 60 \end{vmatrix} \\ &= 40(1 \times 60 - 1 \times 12) \\ &= 40(60 - 12) \\ &= 40 \times 48 \\ &= 1920\end{aligned}$$

1 d)

$$\begin{aligned}&480 \begin{vmatrix} 1 & 1 \\ 1 & 5 \end{vmatrix} \\ &= 480(1 \times 5 - 1 \times 1) \\ &= 480(5 - 1) \\ &= 480 \times 4 \\ &= 1920\end{aligned}$$

2 a)

$$
\begin{aligned}
|\mathbf{A}| &= \begin{vmatrix} 120 & 21 \\ 150 & 35 \end{vmatrix} \\
&= 120 \times 35 - 21 \times 150 \\
&= 4200 - 3150 \\
&= 1050
\end{aligned}
$$

2 b)

$$
\begin{aligned}
&2\begin{vmatrix} 60 & 21 \\ 75 & 35 \end{vmatrix} \\
&= 2(60 \times 35 - 21 \times 75) \\
&= 2(2100 - 1575) \\
&= 2 \times 525 \\
&= 1050
\end{aligned}
$$

2 c)

$$
\begin{aligned}
&6\begin{vmatrix} 20 & 21 \\ 25 & 35 \end{vmatrix} \\
&= 6(20 \times 35 - 21 \times 25) \\
&= 6(700 - 525) \\
&= 6 \times 175 \\
&= 1050
\end{aligned}
$$

2 d)

$$
\begin{aligned}
&30\begin{vmatrix} 4 & 21 \\ 5 & 35 \end{vmatrix} \\
&= 30(4 \times 35 - 21 \times 5) \\
&= 30(140 - 125) \\
&= 30 \times 15 \\
&= 1050
\end{aligned}
$$

2 e)

$$
\begin{aligned}
&150\begin{vmatrix} 4 & 21 \\ 1 & 7 \end{vmatrix} \\
&= 150(4 \times 7 - 21 \times 1) \\
&= 150(28 - 21) \\
&= 150 \times 7 \\
&= 1050
\end{aligned}
$$

2 f)

$$
\begin{aligned}
&1050\begin{vmatrix}4 & 3\\ 1 & 1\end{vmatrix}\\
&= 1050(4\times 1 - 3\times 1)\\
&= 1050(4-3)\\
&= 1050\times 1\\
&= 1050
\end{aligned}
$$

2 g)

$$
\begin{aligned}
|\mathbf{A}| &= \begin{vmatrix}120 & 21\\ 150 & 35\end{vmatrix}\\
&= 2\begin{vmatrix}60 & 21\\ 75 & 35\end{vmatrix}\\
&= 2\times 3\begin{vmatrix}20 & 21\\ 25 & 35\end{vmatrix}\\
&= 2\times 3\times 5\begin{vmatrix}4 & 21\\ 5 & 35\end{vmatrix}\\
&= 2\times 3\times 5\times 5\begin{vmatrix}4 & 21\\ 1 & 7\end{vmatrix}\\
&= 2\times 3\times 5\times 5\times 7\begin{vmatrix}4 & 3\\ 1 & 1\end{vmatrix}\\
&= 2\times 3\times 5\times 5\times 7\times 1\\
\therefore |\mathbf{A}| &= 1050 = 2\times 3\times 5^2\times 7
\end{aligned}
$$

3 a)

$$
\begin{aligned}
|\mathbf{A}| &= \begin{vmatrix}k & k+1\\ 2k & 4(k+1)\end{vmatrix}\\
&= k\times 4(k+1) - (k+1)\times 2k\\
&= 4k^2 + 4k - 2k^2 - 2k\\
&= 2k^2 + 2k\\
&= 2k(k+1)
\end{aligned}
$$

3 b)

$$
\begin{aligned}
& k\begin{vmatrix} 1 & k+1 \\ 2 & 4(k+1) \end{vmatrix} \\
&= k(1 \times 4(k+1) - (k+1) \times 2) \\
&= k(4k + 4 - 2k - 2) \\
&= k(2k + 2) \\
&= 2k(k+1)
\end{aligned}
$$

3 c)

$$
\begin{aligned}
& 2k\begin{vmatrix} 1 & k+1 \\ 1 & 2(k+1) \end{vmatrix} \\
&= 2k(1 \times 2(k+1) - (k+1) \times 1) \\
&= 2k(2k + 2 - k - 1) \\
&= 2k(k+1)
\end{aligned}
$$

3 d)

$$
\begin{aligned}
& 2k(k+1)\begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix} \\
&= 2k(k+1)(1 \times 2 - 1 \times 1) \\
&= 2k(k+1) \times 1 \\
&= 2k(k+1)
\end{aligned}
$$

3 e)

$$|\mathbf{A}| = 2k(k+1)$$

4 a)

$$
\begin{aligned}
|\mathbf{A}| &= \begin{vmatrix} 18k-9 & 3-k \\ 6k-3 & 2k-6 \end{vmatrix} \\
&= (18k-9) \times (2k-6) - (3-k) \times (6k-3) \\
&= 36k^2 - 108k - 18k + 54 - 18k + 9 + 6k^2 - 3k \\
&= 42k^2 - 147k + 63 \\
&= 21(2k-1)(k-3)
\end{aligned}
$$

4 b)

$$
\begin{aligned}
&(2k-1)\begin{vmatrix} 9 & 3-k \\ 3 & 2k-6 \end{vmatrix} \\
&= (2k-1)(9 \times (2k-6) - (3-k) \times 3) \\
&= (2k-1)(18k - 54 - 9 + 3k) \\
&= (2k-1)(21k - 63) \\
&= 21(2k-1)(k-3)
\end{aligned}
$$

4 c)

$$
\begin{aligned}
&3(2k-1)\begin{vmatrix} 3 & 3-k \\ 1 & 2k-6 \end{vmatrix} \\
&= 3(2k-1)(3 \times (2k-6) - (3-k) \times 1) \\
&= 3(2k-1)(6k - 18 - 3 + k) \\
&= 3(2k-1)(7k - 21) \\
&= 21(2k-1)(k-3)
\end{aligned}
$$

4 d)

$$
\begin{aligned}
&3(2k-1)(3-k)\begin{vmatrix} 3 & 1 \\ 1 & -2 \end{vmatrix} \\
&= 3(2k-1)(3-k)(3 \times (-2) - 1 \times 1) \\
&= 3(2k-1)(3-k) \times (-7) \\
&= -21(2k-1)(3-k) \\
&= 21(2k-1)(k-3)
\end{aligned}
$$

5)

$$|\mathbf{A}| = \begin{vmatrix} 4k & 6k-2 \\ 20k & 9k-3 \end{vmatrix}$$

Factor $4k$ out of C1:

$$= 4k \begin{vmatrix} 1 & 2(3k-1) \\ 5 & 3(3k-1) \end{vmatrix}$$

Factor $(3k-1)$ out of C2:

$$= 4k(3k-1) \begin{vmatrix} 1 & 2 \\ 5 & 3 \end{vmatrix}$$

Then:

$$\begin{aligned} &= 4k(3k-1)(1 \times 3 - 2 \times 5) \\ &= 4k(3k-1)(3-10) \\ &= 4k(3k-1) \times -7 \\ &= -28k(3k-1) \end{aligned}$$

6)

$$\begin{aligned} |\mathbf{A}| = 0 \Rightarrow &\begin{vmatrix} 4-8k & 5-10k \\ 3k-6 & 4k-8 \end{vmatrix} = 0 \\ \Rightarrow &\begin{vmatrix} 4(1-2k) & 5(1-2k) \\ 3(k-2) & 4(k-2) \end{vmatrix} = 0 \\ \Rightarrow &(1-2k) \begin{vmatrix} 4 & 5 \\ 3(k-2) & 4(k-2) \end{vmatrix} = 0 \\ \Rightarrow &(1-2k)(k-2) \begin{vmatrix} 4 & 5 \\ 3 & 4 \end{vmatrix} = 0 \\ \Rightarrow &(1-2k)(k-2)(4 \times 4 - 5 \times 3) = 0 \\ \Rightarrow &(1-2k)(k-2) = 0 \\ \Rightarrow &k = \frac{1}{2} \text{ or } k = 2 \end{aligned}$$

7 a)

$$|\mathbf{A}| = \begin{vmatrix} 10 & 8 \\ 8 & 8 \end{vmatrix} = 10 \times 8 - 8 \times 8 = 80 - 64 = 16$$

7 b)

$$|\mathbf{A}| = \begin{vmatrix} 10 & 8 \\ 8 & 8 \end{vmatrix}$$

Factor 2 out of R1:

$$|\mathbf{A}| = 2\begin{vmatrix} 5 & 4 \\ 8 & 8 \end{vmatrix}$$

Factor 8 out of R2:

$$|\mathbf{A}| = 16\begin{vmatrix} 5 & 4 \\ 1 & 1 \end{vmatrix}$$

Let the new R1 = R1 − 4R2:

$$|\mathbf{A}| = 16\begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix}$$

7 c)

$$|\mathbf{A}| = 16\begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix} = 16 \times (1 \times 1 - 0 \times 1) = 16 \times 1 = 16$$

8 a)

$$|\mathbf{A}| = \begin{vmatrix} 16 & 20 \\ 9 & 5 \end{vmatrix} = 16 \times 5 - 20 \times 9 = 80 - 180 = -100$$

8 b)

$$|\mathbf{A}| = \begin{vmatrix} 16 & 20 \\ 9 & 5 \end{vmatrix}$$

Factor 5 out of C2:

$$|\mathbf{A}| = 5\begin{vmatrix} 16 & 4 \\ 9 & 1 \end{vmatrix}$$

Let the new C1 = C1 − 4C2:

$$|\mathbf{A}| = 5\begin{vmatrix} 0 & 4 \\ 5 & 1 \end{vmatrix}$$

8 c)

$$|\mathbf{A}| = 5\begin{vmatrix} 0 & 4 \\ 5 & 1 \end{vmatrix} = 5 \times (0 \times 1 - 4 \times 5) = 5 \times (-20) = -100$$

9 a)

$$|\mathbf{A}| = \begin{vmatrix} 40 & 35 \\ 55 & 25 \end{vmatrix} = 40 \times 25 - 35 \times 55 = 1000 - 1925 = -925$$

9 b)

$$|\mathbf{A}| = \begin{vmatrix} 40 & 35 \\ 55 & 25 \end{vmatrix}$$

Let the new C1 = C1 − 2C2:

$$|\mathbf{A}| = \begin{vmatrix} -30 & 35 \\ 5 & 25 \end{vmatrix}$$

Let the new R1 = R1 + 6R2:

$$|\mathbf{A}| = \begin{vmatrix} 0 & 185 \\ 5 & 25 \end{vmatrix}$$

9 c)

$$|\mathbf{A}| = \begin{vmatrix} 0 & 185 \\ 5 & 25 \end{vmatrix} = 0 \times 25 - 185 \times 5 = -925$$

10 a)

$$|\mathbf{A}| = \begin{vmatrix} 4k+5 & 53 \\ k & 12 \end{vmatrix} = (4k+5) \times 12 - 53 \times k = 48k + 60 - 53k = 60 - 5k$$

10 b)

$$|\mathbf{A}| = \begin{vmatrix} 4k+5 & 53 \\ k & 12 \end{vmatrix}$$

Let the new R1 = R1 − 4R2:

$$|\mathbf{A}| = \begin{vmatrix} 5 & 5 \\ k & 12 \end{vmatrix}$$

Let the new C2 = C2 − C1:

$$|\mathbf{A}| = \begin{vmatrix} 5 & 0 \\ k & 12-k \end{vmatrix}$$

10 c)

$$|\mathbf{A}| = \begin{vmatrix} 5 & 0 \\ k & 12-k \end{vmatrix} = 5 \times (12-k) - 0 \times k = 60 - 5k$$

11 a)

Grace has made a mistake at step (3), where she has attempted to factor 3 out of column 1. 1 is the top-left element in the matrix by stage (2), so if 3 were factored out it would become $\frac{1}{3}$ rather than staying as 1.

11 b)

$$|\mathbf{A}| = \begin{vmatrix} q+1 & -5p \\ 3q+3 & 15p \end{vmatrix} = \begin{vmatrix} q+1 & -5p \\ 3(q+1) & 15p \end{vmatrix}$$

Factor $(q+1)$ out of C1:

$$= (q+1)\begin{vmatrix} 1 & -5p \\ 3 & 15p \end{vmatrix}$$

Factor $5p$ out of C2:

$$= 5p(q+1)\begin{vmatrix} 1 & -1 \\ 3 & 3 \end{vmatrix}$$

Let the new C1 = C1 − C2:

$$= 5p(q+1)\begin{vmatrix} 2 & -1 \\ 0 & 3 \end{vmatrix}$$

Then:

$$= 30p(q+1)$$

12 a)

$$\begin{vmatrix} x & y \\ x^2 & y^2 \end{vmatrix}$$

Factor x out of C1:

$$= x \begin{vmatrix} 1 & y \\ x & y^2 \end{vmatrix}$$

Factor y out of C2:

$$= xy \begin{vmatrix} 1 & 1 \\ x & y \end{vmatrix}$$

Let the new C2 = C2 − C1:

$$= xy \begin{vmatrix} 1 & 0 \\ x & y - x \end{vmatrix}$$

Then:

$$= xy(y - x)$$

12 b)

$$\begin{vmatrix} x^2 & y^2 \\ xy & xy \end{vmatrix}$$

Factor x out of C1:

$$= x \begin{vmatrix} x & y^2 \\ y & xy \end{vmatrix}$$

Factor y out of C2:

$$= xy \begin{vmatrix} x & y \\ y & x \end{vmatrix}$$

Let the new R1 = R1 + R2:

$$= xy \begin{vmatrix} x+y & x+y \\ y & x \end{vmatrix}$$

Factor $(x + y)$ out of R1:

$$= xy(x+y) \begin{vmatrix} 1 & 1 \\ y & x \end{vmatrix}$$

Let the new C2 = C2 − C1:

$$= xy(x+y) \begin{vmatrix} 1 & 0 \\ y & x-y \end{vmatrix}$$

Then:

$$= xy(x+y)(x-y)$$

12 c)

$$\begin{vmatrix} xy & xy^2 \\ x^2 & xy \end{vmatrix}$$

Factor x out of C1:

$$= x \begin{vmatrix} y & xy^2 \\ x & xy \end{vmatrix}$$

Factor y out of C2:

$$= xy \begin{vmatrix} y & xy \\ x & x \end{vmatrix}$$

Factor y out of R1:

$$= xy^2 \begin{vmatrix} 1 & x \\ x & x \end{vmatrix}$$

Factor x out of R2:

$$= x^2y^2 \begin{vmatrix} 1 & x \\ 1 & 1 \end{vmatrix}$$

Let the new R2 = R2 − R1:

$$= x^2y^2 \begin{vmatrix} 1 & x \\ 0 & 1-x \end{vmatrix}$$

Then:

$$= x^2y^2(1-x)$$

12 d)

$$\begin{vmatrix} yz & xy \\ xyz & xyz \end{vmatrix}$$

Factor y out of R1:

$$= y \begin{vmatrix} z & x \\ xyz & xyz \end{vmatrix}$$

Factor xyz out of R2:

$$= xy^2z \begin{vmatrix} z & x \\ 1 & 1 \end{vmatrix}$$

Let the new C1 = C1 − C2:

$$= xy^2z \begin{vmatrix} z-x & x \\ 0 & 1 \end{vmatrix}$$

Then:

$$= xy^2z(z-x)$$

12 e)

$$\begin{vmatrix} xyz & yz \\ xy & xyz \end{vmatrix}$$

Factor y out of R1:

$$= y \begin{vmatrix} xz & z \\ xy & xyz \end{vmatrix}$$

Factor xy out of R2:

$$= xy^2 \begin{vmatrix} xz & z \\ 1 & z \end{vmatrix}$$

Let the new R1 = R1 − R2:

$$= xy^2 \begin{vmatrix} xz-1 & 0 \\ 1 & z \end{vmatrix}$$

Then:

$$= xy^2z(xz-1)$$

12 f)

$$\begin{vmatrix} x^2y & xz^2 \\ xy^2 & xyz^2 \end{vmatrix}$$

Factor z^2 out of C2:

$$= z^2 \begin{vmatrix} x^2y & x \\ xy^2 & xy \end{vmatrix}$$

Factor xy out of R2:

$$= xyz^2 \begin{vmatrix} x^2y & x \\ y & 1 \end{vmatrix}$$

Factor y out of C1:

$$= xy^2z^2 \begin{vmatrix} x^2 & x \\ 1 & 1 \end{vmatrix}$$

Factor x out of R1:

$$= x^2y^2z^2 \begin{vmatrix} x & 1 \\ 1 & 1 \end{vmatrix}$$

Let the new R1 = R1 − R2:

$$= x^2y^2z^2 \begin{vmatrix} x-1 & 0 \\ 1 & 1 \end{vmatrix}$$

Then:

$$= x^2y^2z^2(x-1)$$

12 g)

$$\begin{vmatrix} x^2y^2z & xy^2z^2 \\ x^2yz^2 & xyz \end{vmatrix}$$

Factor xyz out of R1:

$$= xyz \begin{vmatrix} xy & yz \\ x^2yz^2 & xyz \end{vmatrix}$$

Factor xyz out of R2:

$$= x^2y^2z^2 \begin{vmatrix} xy & yz \\ xz & 1 \end{vmatrix}$$

Factor x out of C1:

$$= x^3y^2z^2 \begin{vmatrix} y & yz \\ z & 1 \end{vmatrix}$$

Factor y out of R1:

$$= x^3y^3z^2 \begin{vmatrix} 1 & z \\ z & 1 \end{vmatrix}$$

Let the new R1 = R1 + R2:

$$= x^3y^3z^2 \begin{vmatrix} 1+z & 1+z \\ z & 1 \end{vmatrix}$$

Factor $(1 + z)$ out of R1:

$$= x^3y^3z^2(1+z) \begin{vmatrix} 1 & 1 \\ z & 1 \end{vmatrix}$$

Let the new R1 = R1 − R2:

$$= x^3y^3z^2(1+z) \begin{vmatrix} 1-z & 0 \\ z & 1 \end{vmatrix}$$

Then:

$$= x^3y^3z^2(1+z)(1-z)$$

13 a)

$$\begin{vmatrix} p & -p-2q \\ 4p+5q & p+2q \end{vmatrix} = \begin{vmatrix} p & -(p+2q) \\ 4p+5q & p+2q \end{vmatrix}$$

Factor $(p+2q)$ out of C1:

$$= (p+2q)\begin{vmatrix} p & -1 \\ 4p+5q & 1 \end{vmatrix}$$

Let the new R1 = R1 + R2:

$$= (p+2q)\begin{vmatrix} 5p+5q & 0 \\ 4p+5q & 1 \end{vmatrix} = (p+2q)\begin{vmatrix} 5(p+q) & 0 \\ 4(p+q) & 1 \end{vmatrix}$$

Factor $(p+q)$ out of C1:

$$= (p+2q)(p+q)\begin{vmatrix} 5 & 0 \\ 4 & 1 \end{vmatrix}$$

Then:

$$= 5(p+2q)(p+q)$$

13 b)

$$\begin{vmatrix} 2q-2p & 3p-3q \\ 14q-6p & 2p \end{vmatrix} = \begin{vmatrix} -2(p-q) & 3(p-q) \\ 14q-6p & 2p \end{vmatrix}$$

Factor $(p-q)$ out of R1:

$$= (p-q)\begin{vmatrix} -2 & 3 \\ 14q-6p & 2p \end{vmatrix}$$

Factor 2 out of R2:

$$= 2(p-q)\begin{vmatrix} -2 & 3 \\ 7q-3p & p \end{vmatrix}$$

Let the new C1 = C1 + 3C2:

$$= 2(p-q)\begin{vmatrix} 7 & 3 \\ 7q & p \end{vmatrix}$$

Factor 7 out of C1:

$$= 14(p-q)\begin{vmatrix} 1 & 3 \\ q & p \end{vmatrix}$$

Let the new C2 = C2 − 3C1:

$$= 14(p-q)\begin{vmatrix} 1 & 0 \\ q & p-3q \end{vmatrix}$$

Then:

$$= 14(p-q)(p-3q)$$

13 c)

$$\begin{vmatrix} 16p+5q-13 & 4-8p \\ 6q-12p & 6p-3 \end{vmatrix} = \begin{vmatrix} 16p+5q-13 & -4(2p-1) \\ 6q-12p & 3(2p-1) \end{vmatrix}$$

Factor $(2p-1)$ out of C1:

$$= (2p-1)\begin{vmatrix} 16p+5q-13 & -4 \\ 6q-12p & 3 \end{vmatrix}$$

Let the new R1 = R1 + $\frac{4}{3}$ × R2:

$$= (2p-1)\begin{vmatrix} 13q-13 & 0 \\ 6q-12p & 3 \end{vmatrix}$$

Factor $13(q-1)$ out of R1:

$$= 13(2p-1)(q-1)\begin{vmatrix} 1 & 0 \\ 6q-12p & 3 \end{vmatrix}$$

Then:

$$= 39(2p-1)(q-1)$$

13 d)

$$\begin{vmatrix} 3-2q & 4q-8p \\ 6-4q & 8q+24p-30 \end{vmatrix} = \begin{vmatrix} 3-2q & 4q-8p \\ 2(3-2q) & 8q+24p-30 \end{vmatrix}$$

Factor $(3-2q)$ out of C1:

$$= (3-2q)\begin{vmatrix} 1 & 4q-8p \\ 2 & 8q+24p-30 \end{vmatrix}$$

Let the new R2 = R2 − 2R1:

$$= (3-2q)\begin{vmatrix} 1 & 4q-8p \\ 0 & 40p-30 \end{vmatrix}$$

Factor $10(4p-3)$ out of R2:

$$= 10(3-2q)(4p-3)\begin{vmatrix} 1 & 4q-8p \\ 0 & 1 \end{vmatrix}$$

Then:

$$= 10(3-2q)(4p-3)$$

14)

$$|\mathbf{A}| = xy\begin{vmatrix} 1 & -3 \\ 2 & 4 \end{vmatrix}$$

Multiply C1 by x:

$$= y\begin{vmatrix} x & -3 \\ 2x & 4 \end{vmatrix}$$

Multiply C2 by y:

$$= \begin{vmatrix} x & -3y \\ 2x & 4y \end{vmatrix}$$

Let the new R2 = R2 − 2R1:

$$= \begin{vmatrix} x & -3y \\ 0 & 10y \end{vmatrix} \quad \text{as required}$$

15)

$$|\mathbf{A}| = x(x-1)y\begin{vmatrix}2 & -3\\ 3 & -1\end{vmatrix}$$

Multiply C1 by x:

$$= (x-1)y\begin{vmatrix}2x & -3\\ 3x & -1\end{vmatrix}$$

Multiply C2 by $(x-1)y$:

$$= \begin{vmatrix}2x & -3(x-1)y\\ 3x & -(x-1)y\end{vmatrix} = \begin{vmatrix}2x & 3y-3xy\\ 3x & y-xy\end{vmatrix}$$

Multiply R1 by 3:

$$= \frac{1}{3}\begin{vmatrix}6x & 9y-9xy\\ 3x & y-xy\end{vmatrix}$$

Multiply R2 by 2:

$$= \frac{1}{6}\begin{vmatrix}6x & 9y-9xy\\ 6x & 2y-2xy\end{vmatrix}$$

Factor 6 out of C1:

$$= \begin{vmatrix}x & 9y-9xy\\ x & 2y-2xy\end{vmatrix}$$

Let the new R1 = R1 − R2:

$$= \begin{vmatrix}0 & 7y-7xy\\ x & 2y-2xy\end{vmatrix}$$

Let the new C2 = C2 + C1:

$$= \begin{vmatrix}0 & 7y-7xy\\ x & x+2y-2xy\end{vmatrix} \quad \text{as required}$$

16)

$\mathbf{A}$ is singular $\Rightarrow |\mathbf{A}| = 0$

$$\begin{vmatrix} x-5a & -4a \\ 14x+2a & 9x+7a \end{vmatrix} = 0$$

Multiply C1 by 4:

$$\Rightarrow \frac{1}{4}\begin{vmatrix} 4x-20a & -4a \\ 56x+8a & 9x+7a \end{vmatrix} = 0$$

Let the new C1 = C1 − 5C2:

$$\Rightarrow \frac{1}{4}\begin{vmatrix} 4x & -4a \\ 11x-27a & 9x+7a \end{vmatrix} = 0$$

Let the new C2 = C2 + C1:

$$\Rightarrow \frac{1}{4}\begin{vmatrix} 4x & 4x-4a \\ 11x-27a & 20x-20a \end{vmatrix} = 0$$

$$\Rightarrow \frac{1}{4}\begin{vmatrix} 4x & 4(x-a) \\ 11x-27a & 20(x-a) \end{vmatrix} = 0$$

Factor $(x-a)$ out of C2:

$$\Rightarrow \frac{1}{4}(x-a)\begin{vmatrix} 4x & 4 \\ 11x-27a & 20 \end{vmatrix} = 0$$

Multiply R1 by 5:

$$\Rightarrow \frac{1}{20}(x-a)\begin{vmatrix} 20x & 20 \\ 11x-27a & 20 \end{vmatrix} = 0$$

Let the new R1 = R1 − R2:

$$\Rightarrow \frac{1}{20}(x-a)\begin{vmatrix} 9x+27a & 0 \\ 11x-27a & 20 \end{vmatrix} = 0$$

Then:

$$\Rightarrow \frac{1}{20}(x-a) \times (9x+27a) \times 20 = 0$$
$$\Rightarrow 9(x-a)(x+3a) = 0$$
$$\Rightarrow (x-a)(x+3a) = 0$$
$$\therefore x = a \text{ or } x = -3a$$

17)

$\mathbf{A}$ is singular $\Rightarrow |\mathbf{A}| = 0$

$$\begin{vmatrix} 8a^2 - 2x^2 & 3x^2 - 6ax - 24a^2 \\ 5ax - 10a^2 & 8ax - 32a^2 \end{vmatrix} = 0$$
$$\Rightarrow \begin{vmatrix} 2(4a^2 - x^2) & 3(x^2 - 2ax - 8a^2) \\ 5a(x - 2a) & 8a(x - 4a) \end{vmatrix} = 0$$
$$\Rightarrow \begin{vmatrix} -2(x - 2a)(x + 2a) & 3(x - 4a)(x + 2a) \\ 5a(x - 2a) & 8a(x - 4a) \end{vmatrix} = 0$$

Factor a out of R2:

$$\Rightarrow a \begin{vmatrix} -2(x - 2a)(x + 2a) & 3(x - 4a)(x + 2a) \\ 5(x - 2a) & 8(x - 4a) \end{vmatrix} = 0$$

Divide through by a:

$$\Rightarrow \begin{vmatrix} -2(x - 2a)(x + 2a) & 3(x - 4a)(x + 2a) \\ 5(x - 2a) & 8(x - 4a) \end{vmatrix} = 0$$

Factor $(x - 2a)$ out of C1:

$$\Rightarrow (x - 2a) \begin{vmatrix} -2(x + 2a) & 3(x - 4a)(x + 2a) \\ 5 & 8(x - 4a) \end{vmatrix} = 0$$

Factor $(x + 2a)$ out of R1:

$$\Rightarrow (x + 2a)(x - 2a) \begin{vmatrix} -2 & 3(x - 4a) \\ 5 & 8(x - 4a) \end{vmatrix} = 0$$

Factor $(x - 4a)$ out of C2:

$$\Rightarrow (x + 2a)(x - 2a)(x - 4a) \begin{vmatrix} -2 & 3 \\ 5 & 8 \end{vmatrix} = 0$$

Divide through by $\begin{vmatrix} -2 & 3 \\ 5 & 8 \end{vmatrix}$ (as it is just a constant):

$$\Rightarrow (x + 2a)(x - 2a)(x - 4a) = 0$$
$$\therefore x = -2a \text{ or } x = 2a \text{ or } x = 4a$$

18)

$$\begin{vmatrix} 3a-4b-3c & a-2b-2c \\ -a+4b-c & a+2b-2c \end{vmatrix}$$

Let the new R1 = R1 + R2:

$$= \begin{vmatrix} 2a-4c & 2a-4c \\ -a+4b-c & a+2b-2c \end{vmatrix}$$

Factor $(2a-4c)$ out of R1:

$$= (2a-4c)\begin{vmatrix} 1 & 1 \\ -a+4b-c & a+2b-2c \end{vmatrix}$$

Let the new C2 = C2 − C1:

$$= (2a-4c)\begin{vmatrix} 1 & 0 \\ -a+4b-c & 2a-2b-c \end{vmatrix}$$

Then:

$$= (2a-4c)(2a-2b-c)$$

$$= 2(a-2c)(2a-2b-c)$$

19)

$$\begin{vmatrix} pq-4p+5q-20 & 3pq+15q \\ 3pq-12p-6q+24 & 4pq-10p-8q+20 \end{vmatrix}$$
$$= \begin{vmatrix} (p+5)(q-4) & 3q(p+5) \\ 3(pq-4p-2q+8) & 4pq-10p-8q+20 \end{vmatrix}$$
$$= \begin{vmatrix} (p+5)(q-4) & 3q(p+5) \\ 3(p-2)(q-4) & 4pq-10p-8q+20 \end{vmatrix}$$

Factor $(q-4)$ out of C1:

$$= (q-4)\begin{vmatrix} p+5 & 3q(p+5) \\ 3(p-2) & 4pq-10p-8q+20 \end{vmatrix}$$

Factor $(p+5)$ out of R1:

$$= (p+5)(q-4)\begin{vmatrix} 1 & 3q \\ 3p-6 & 4pq-10p-8q+20 \end{vmatrix}$$

Let the new C2 = C2 + 6C1:

$$= (p+5)(q-4)\begin{vmatrix} 1 & 3q+6 \\ 3p-6 & 4pq+8p-8q-16 \end{vmatrix}$$

$$= (p+5)(q-4)\begin{vmatrix} 1 & 3(q+2) \\ 3(p-2) & 4(pq+2p-2q-4) \end{vmatrix}$$

$$= (p+5)(q-4)\begin{vmatrix} 1 & 3(q+2) \\ 3(p-2) & 4(p-2)(q+2) \end{vmatrix}$$

Factor $(p-2)$ out of R2:

$$= (p-2)(p+5)(q-4)\begin{vmatrix} 1 & 3(q+2) \\ 3 & 4(q+2) \end{vmatrix}$$

Factor $(q+2)$ out of C2:

$$= (p-2)(p+5)(q+2)(q-4)\begin{vmatrix} 1 & 3 \\ 3 & 4 \end{vmatrix}$$

Let the new R2 = R2 − 3R1:

$$= (p-2)(p+5)(q+2)(q-4)\begin{vmatrix} 1 & 3 \\ 0 & -5 \end{vmatrix}$$

Then:

$$= -5(p-2)(p+5)(q+2)(q-4)$$

20 a)

$$\begin{vmatrix} x+3a+b & x+3a \\ 5a+b & 5a \end{vmatrix}$$

Let the new C1 = C1 − C2:

$$= \begin{vmatrix} b & x+3a \\ b & 5a \end{vmatrix}$$

Factor b out of C1:

$$= b \begin{vmatrix} 1 & x+3a \\ 1 & 5a \end{vmatrix}$$

Let the new R2 = R2 − R1:

$$= b \begin{vmatrix} 1 & x+3a \\ 0 & 2a-x \end{vmatrix}$$

Then:

$$= b(2a-x)$$

20 b)

$$\begin{aligned} 5a = 50 &\Rightarrow a = 10 \\ 50 + b = 55 &\Rightarrow b = 5 \\ |\mathbf{A}| = \pm\frac{50}{20} &= \pm\frac{5}{2} \end{aligned}$$

If $|\mathbf{A}| = \frac{5}{2}$:

$$\begin{aligned} &\Rightarrow 5(2 \times 10 - x) = \frac{5}{2} \\ &\Rightarrow 20 - x = \frac{1}{2} \\ &\Rightarrow x = \frac{39}{2} \end{aligned}$$

If $|\mathbf{A}| = -\frac{5}{2}$:

$$\begin{aligned} &\Rightarrow 5(2 \times 10 - x) = -\frac{5}{2} \\ &\Rightarrow 20 - x = -\frac{1}{2} \\ &\Rightarrow x = \frac{41}{2} \\ &\therefore x = \frac{39}{2} \text{ or } x = \frac{41}{2} \end{aligned}$$